

So, you've decided to DxNALE?

Preface

Considerations

- Are you capable of scouring over wiring diagrams?
 - This is not a step-by-step guide, it's more high level instructions and descriptions of what is required. I think I've done a good job of documenting most of the important stuff—But I absolutely may have missed something.
- Do you really want to do this in the first place?
 - Playing DX is nice, but it is a ton of effort. Also, you won't be able to return to FiNALE easily.
 - This is still cost effective compared to real DX, but again, work. Lots of it. Be prepared to sacrifice a week+ of your time.
- Is there anything you need to replace on your cab?
 - You should try to get your cab running FiNALE perfectly before doing this jump. It'll help investigating issues later.
 - It'll save potential work items and purchase items if it all works.
- Do you care about getting on Mythos?
 - You don't need a pi if not. You can just set up serial for your ADX cons, and run the data USB directly to your PC.
- Do you care about player cameras/code readers?
 - You can save some money and effort running cables here. If you're mythos-bound, Felix can disable this functionality for you.
- Do you care about LEDs?
 - You can save some effort here if your LED controllers were dead, or you don't care about the billboard LEDs.

Costs

To personally detail the costs on my own cab,

- Cab - \$3250
- Shipping – 1500\$ (total \$4750)
- Controllers – 2000\$ + 300\$ shipping to US (total \$7050)
- [Rabbit buttons](#) – 440\$ + 30\$ shipping (total \$7520)
- [Raspberry Pi 5](#) (I got a shitty starter kit because I'm lazy) – 130\$. (total \$7560)

- [Headphone amps](#) – 30\$ a pop, 60\$ total (total \$7710)
- [FTDI USB-to-serial cables](#) – 14\$ a pop, 56\$ total for 4, 2 touch, 2 LED. (total \$7766)
- JST [SM](#) and [XH](#) connector/crimp kit – 38\$ total (total \$7804)
- Star Horse Aime reader/VFD combo – 41\$ + 27\$ shipping (total \$7872)
- PLA filament; under 1kg is actually needed to print the right parts; I just used more designing. You can get black PLA filament pretty ez for ~20-40\$. Let's call that 20\$. (total \$7892)
- ALLS MX2 (HX2 ideally; if you don't care about Mythos, this matters infinitely less. You can downgrade to HX2 spec.) and IO4 board – I traded my og Wacca LEDs and 300\$ for this. Let's call that 300\$. (total \$8192)
- [DP-DVI cable](#)– 10\$ (total \$8202)
- DVI-D cable, the cab doesn't come with one, the monitors work fine with it. RE used DVI-I. [Also easily available for \\$10.](#) (total \$8212)
- USB hub, preferably powered, or a sufficient power supply for your pi. I got mine for way too much honestly, get a cheaper one. 40\$. (total \$8252).
- A second USB hub for your ALLS if you plan on using the player camera and code readers.
- [Player camera](#) – 15\$. (total \$8267)
- [Code reader cameras](#) – 15\$. (total \$8282)
- Fresh display. Almost 300\$ after shipping. We'll call that \$8582.
- [14 pin molex connector](#) for power. 14\$. (total \$8596)

Work Items

Mandatory

Touch screens and ADX PCBs



The touch assemblies are straightforward. 8 screws hold on the glass panel. Swap that sucker, wire it up to the ADX PCBs. I recommend mounting the sensor PCB to the back of the glass, just atop the ribbon cable (but not to the ribbon cable!) using 3m command strips. I also mounted the ADX PCB to the bottom of the touch assy.

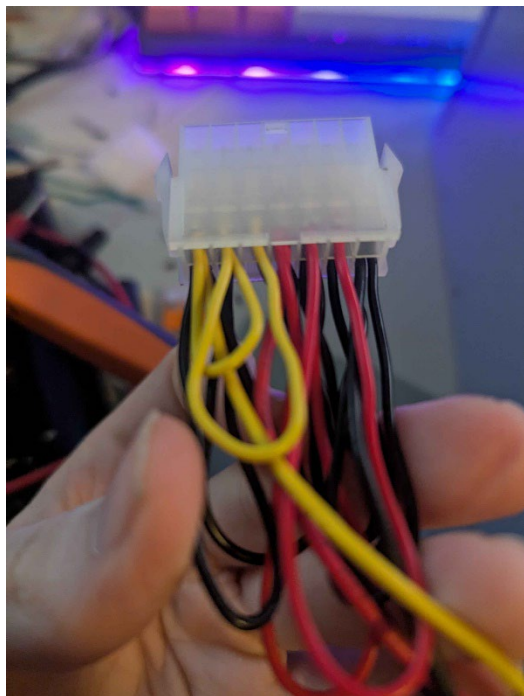
P1 touch lives on COM3, P2 touch lives on COM4.

LEDs, P1 lives on COM21, P2 on COM23.

Mythos-compatible setup will be described later. If running data, just plug the data USBs into the ALLS.

Computer replacement

Ringedge 2 can't run DX. To get on Mythos, you'll want an ALLS HX2; [MX2 can be downgraded to HX2](#). You'll need primary storage (256gb) and substorage (512gb) if you're wanting to get on Mythos. Don't worry about this for data.

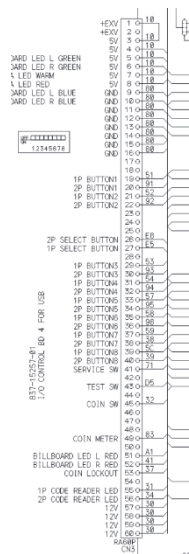


A big concern here is now power. You'll want to either plug into a Molex connector to get 5v and 12v, or splice into the cable.

or you could get the ringedge's PSU booting with the ALLS. Don't do this. This is stupid. The two barely fit in the chassis.

The top pins are 3 12v, 3 5v, and 1 3.3v pin. You don't need to worry about the 3.3v pin for this. Don't bother. It isn't used.

IO Board

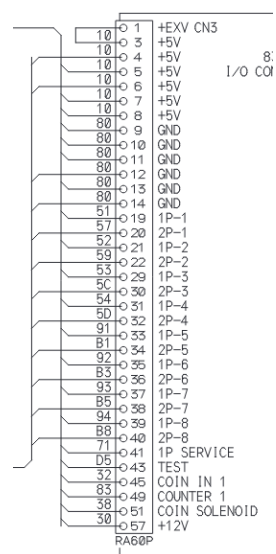


Maimai DX IO Board 60 pin

An IO4 board is mandatory to run DX. Most of the wiring is the same, save for three parts.

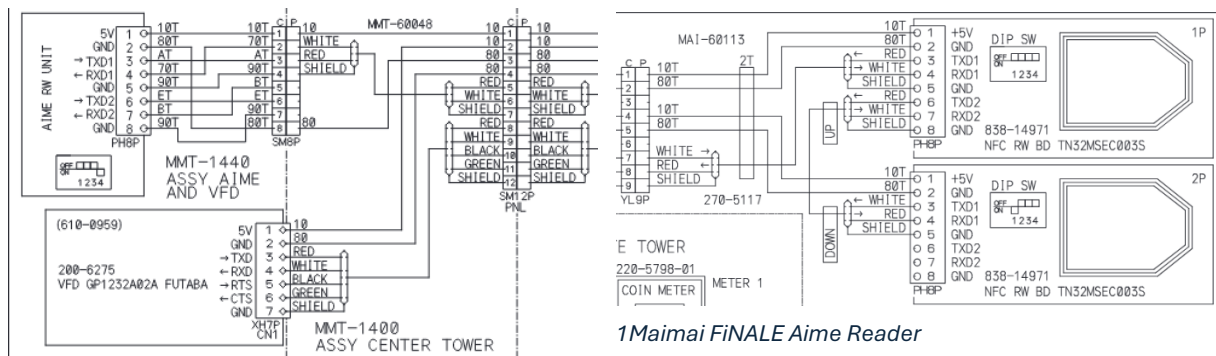
The coin blocker pin is moved to pin 53 from 51, and in its place, P1 and P2 select buttons are wired in. These are mandatory for sorting. Depin and move the coin blocker, wire in the P1/2 buttons.

We'll discuss the other two parts in further detail later, as they are optional additions.



Maimai FiNALE IO Board 60 pin

Aime Reader (Optionally, VFD)



Maimai DX Aime Reader Wiring

You cannot re-use the card readers originally installed in the cab. You'll need to attach a modern, gen 3 Aime reader (Part # 610-0955). The pinout is remarkably the same. You should be able to re-use the serial connection here for at least the reader if you wish. ***! WOULD STRONGLY ADVISE VALIDATING MY CLAIMS.***

Personally I wired up a new connector for both the VFD and Aime reader, and borrowed 5 volts from the original connectors. These belong on COM1 (Aime Reader), and COM2 (VFD). I also re-used the 5v for lighting up my P1/2 buttons, but that is not mandatory.

Display Connections

P1 should be connected via displayport. The manual lies and says HDMI. P2 should be wired up via DVI. HDMI is valid to install recording equipment on. I'm not sure how to record the other player, but I assume a second DP connection?

Non-Mandatory

Headphone support

Pick or build your favorite headphone amps of choice. The ALLS does not output significantly enough amplified audio to drive even cheapo earbuds on its own. There are then, three ways to get headphones out of the cab.

- 1) Hide the amps inside the cab. Drill holes in the bottom cable covers to expose a female audio cable or jack to the player. This option is destructive, but the least jank option.
- 2) Non-destructive, run a flat headphone cable with a female end out the same area from under the shell, instead of out the cable cover. This will reduce the hassle of

putting the touch assemblies back on, but incurs a risk of breaking your shell in extenuating circumstances.

- 3) Remove the coin selector entirely, and place the headphone amps inside the service door. High jank, low effort.

Center is P1 headphone out, Side is P2 headphone out. Front and rear remain the same between Finale and DX.

Controller LED replacements

If you already had working 15070 LED controllers, I **think** it's fine to just blindly re-use them from FiNALE. They **may** report BAD due to board #, but otherwise will receive data for the button LEDs. You **can** [proxy them on the pi if you wish](#) to make them read GOOD. Some cabs that have been repaired have swapped BLUE/GREEN lines on their LEDs and may output incorrectly without the patch in the proxy.

Allegedly, FET data doesn't get sent if they read BAD. You'll also want to consider separating the side and bottom LEDs on the FET (but this is not necessary), since they are controlled separately on DX.

If your boards were bad like mine though, you'll have to proxy these to your ADX PCBs and insert Yuan's LED PCBs into your buttons.

VFD

This is only really necessary if you want your cab to *eventually* support e-money. For home users and freeplay arcades, this isn't hugely necessary. There are no features in maimai dx, aside from the paid tickets that require "real" money. Worst case, pop open your coin door and flick the switch, it's at the bottom of the coin chute. Posterity item.

Roof/Woofer LEDs

Posterity work item. The IO4 board is wired up for the "billboard" R, G, and B lines on P1 and P2 sides in DX. These are the same 5050 LEDs between DX and FiNALE. On FiNALE, there is also "Center" LEDs, wired up on P1's LED controller. (These are illustrated as "ASSY SW BASE" in the wiring diagram.

Unplug these from the original connector exiting ASSY SW BASE on both sides, create a connector to the IO4 board. For posterity, wire center up to the P1 LEDs (or don't, I'm not your mom. I personally left center unplugged on my cab.)

Code Readers/Player camera

For the player camera, pick up a shitty cheapo camera off amazon. Reverse the cover for the center LED. Get a sturdy mount, stick that sucker on that acrylic. Do not get too fancy of a camera. The game may detect it, but not really use it. I used [this mount](#). +7\$ to my total.

For the code readers, the game will not accept anything other than 640x480 UVC cameras. These need to be aimed at two QR codes that read SDEZ01 and SDEZ02 for P1 and P2 code readers respectively. My model has slots for these. Cards are standard playing card sized (think Weiss Schwarz or Magic; not Yu-gi-oh.) with the QR code in the bottom right. Posterity item. Unless you have a card maker. Even then. Posterity item. Mythos already gives every player a DX pass default.

Code reader LEDs

The game expects to be able to illuminate the code reader LEDs to read codes. These are 5050 LEDs, with R, G, and B wired up to a single pin for P1 and P2. Feed 12v in off a line of your choice. There isn't already 12v available in this area of the cab, so you'll need a new line.

Do not point the LEDs directly at your SDEZ01/SDEZ02 codes. This will make the camera fail to read them. Instead, point them at the cab, and the light will bounce to them.

Mythos Compatibility

You can, naturally, just plug the data lines for the ADX cons into the ALLS and run data on whatever server you please. But these cabs are allowed on Mythos, *provided the game boots without modifications*. Felix can disable the cameras on your cab, but there is no exception for the touch panels. Those bad boys have gotta be plugged into the ALLS via serial.

For this, similarly to the LEDs, a little linux know-how will be required, as well as setting up the proxies. If you don't need the LED boards, the Pi 5 has enough USBs on board to not need a hub. Two data lines, two USB-Serial adaptors. You will probably need null terminal adaptors in between, to swap the Rx/D/TxD lines.

4ndr3w's touch proxy software lives [here](#). This leads nicely to the next section.

Proxy Setup

If you need to run your ADX controllers natively, especially for mythos compatibility, you'll need the serial proxies present. Basic linux knowledge, and/or the ability to read manpages is assumed. The high level steps are as follows:

- Connect USB devices one-by-one, sussing out which is which between a combo of /dev/serial/by-id (Touch and usb-serial) and /dev/serial/by-path (LEDs, since they present identically). Symlink them to a path of choice with a sensible naming scheme (e.g. P1-TOUCH-ALLS, P1-TOUCH-ADX, P1-LED-ALLS, P1-LED-ADX)
- Install rustup, and the rust SDK. Build the proxies.
- Fire them up for the first time, validate your implementation. Touch proxies will be “./maitouch ALLS ADX”, LED will be “./mailight ALLS ADX”.
- Create systemd services. Set to restart on failure. The LEDs will recover fine if they disappear momentarily, the touch panels will not.

Setting up SSH into your pi is reasonable. Do it, in case you need to or want to upgrade this software at some point.